MILITARY SPECIFICATION

LATHE, MEDIUM DUTY, BRAKE DRUM AND DISC BRAKE ROTORS

This specification is approved for use by the U.S. Army Armament Research, Development and Engineering Center, Engineering Support Directorate, Tool and Equipment Engineering Division, Product Engineering Branch, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers a medium duty brake drum and disc brake rotor lathe, pedestal base, floor mounted, with chip catcher and storage board, electrically powered, with equipment and accessories, for turning and truing brake drums and disc brake rotors of commercial and military vehicles.

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-A, Rock Island, IL 61299-7300 by using a self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4910

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SPECIFICATIONS

FEDERAL

FF-B-171	- Bearings, Ball, Annular (General Purpose)
FF-B-185	- Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning
FF-B-187	- Bearing, Roller, Tapered
VV-L-800	- Lubricating Oil, General Purpose, Preservative (Water-Displacing, Low Temperature)
PPP-B-601	- Boxes, Wood, Cleated-Plywood

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MIL-T-152	- Treatment, Moisture and Fungus-Resistant of Communications, Electronic and Associated Electrical Equipment
MIL-C-104	- Crates, Wood: Lumber and Plywood Sheathed, Nailed and Bolted
MIL-M-7298	- Manuals, Technical; Commercial Equipment
MIL-G-10924	- Grease, Automotive and Artillery
MIL-C-16173	- Corrosion Preservative Compound, Solvent Cutback, Cold Application
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In
MIL-B-13501	- Bearings and Bushings, Brass and Bronze, Machined and Formed
MIL-V-173	- Varnish, Moisture and Fungus Resistant

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services
FED-STD-376	- Preferred Metric Units for General Use by the Federal Government

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of US Military Property
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-889	- Dissimilar Metals
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods
MIL-STD-1190	- Minimum Guidelines for Level C Preservation, Packing and Marking
MS19081	- Bearings, Roller, Tapered Single Row of Rollers, Normal Angle

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2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specificatin to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA)

Title 29, of Federal Regulations, Chapter XVII, Part 1910, and amendments, Occupational Safety and Health Administration Standards.

29 CFR 1910 - GENERAL INDUSTRY, OSHA SAFETY AND HEALTH STANDARDS

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA Standards

MG 1 - Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Fire Codes
NFPA 79 - Electrical Standard for Metalworking Machine Tools and Plastics Machinery

(Application for copies should be addessed to the National Fire Protection Association, Batterymarch Park, Quincy MA 02269.)

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AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

360.02	- AGMA Manual for Machine Tool Gearing
390.03-80	- Gear Handbook Volume I: Gear Classification, Materials and Measuring Methods for Unassembled Gears

(Application for copies should be addressed to the American Gear Manufacturers Association, Standards Dept., 1330 Massachusetts Avenue, NW, Washington, DC 20005.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B5.17	- Markings for Identifying Grinding Wheels and Other Bonded Abrasives
B7.1	- American Standard Safety Code for the Use, Care, and Protection of Abrasive Wheels
B46.1	- Surface Texture

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E380-84	- Standard for Metric Practice

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1). The approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of the specification.

3.1.1 Contractor furnished property. Brake drums, brake disc rotors, spindle assemblies, and bearing assemblies necessary to establish conformance with requirements herein shall be provided by the contractor. There will be no Government Furnished Property or Material (GFP/GFM) provided.

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3.2 Design. The medium duty brake drum and disc brake rotor lathe, pedestal base, floor mounted, with chip catcher and storage board, electrically powered, shall include all the features, components, and accessories necessary for turning and truing automotive brake drums and disc brake rotors as specified herein. The lathe shall be new and of the latest design capable of meeting the performance requirements specified herein. The lathe shall be electric motor driven, with automatic and hand feeds. The equipment shall be convenient to operate from the front of the machine, and the operations such as feed selection, tool bar positioning, movement of all slides, (carriage slides, cross slides, etc.) shall be within easy reach of the operator. All parts subject to wear, breakage, or distortion shall be accessible for replacement, adjustment, and repair.

3.2.1 Interchangeability. All parts of the lathe shall be manufactured to definite standards and tolerances that will provide for the interchangeability of respective parts between end items of the same model. All replacement parts shall be interchangeable with their respective original parts, and fit/function with mating components.

3.2.2 Fastening devices. Fastening devices and methods shall be chosen to serve the need while providing necessary adjustability for service, maintenance or repair. Fasteners shall be installed to prevent loss of tightness and shall not loosen in service.

3.2.2.1 Threads. The design, manufacture and configuration of threads shall be examined and inspected to conform with FED-STD-H28 and the applicable detailed standards referenced therein.

3.2.3 Gears. All gear and pinion designs used in the lathe or its components shall be manufactured in accordance with AGMA 390.03, AGMA 360.02 and FED-STD-376. Gears and Mountings shall be designed and machined to provide a fit-up to minimize backlash.

3.2.4 Bearings. Spindles and main drives shall be equipped with tapered roller bearings. Sleeve type spindle bearings are not acceptable. Ball or straight roller bearings shall be in accordance with FF-B-171, FF-B-185, FF-B-187 or MIL-B-13501.

3.2.5 Dials. Dials shall be graduated in the US customary inch and metric (SI) systems of measurement. Dial graduations indicating stock removal or tool movement shall be in increments of not more than 0.001 inch and 0.02mm. Dial graduations shall be easily read from the machine operators normal working position. Graduations shall be permanently and legibly etched or engraved on a contrasting nonglare background. Dials that require more than one revolution to indicate their full range shall be calibrated such that the last dial graduation progresses into and is continuous with the first graduation as the dial is rotated through the zero position for successive revolutions. Dual scale feed adjustment devices shall have independent zero adjustments for the US and Metric graduations on the same devices. Dials shall be accurate indicators for adjustment of tool/component movement.

3.2.6 Lubricating systems. All bearings (except sealed for life, permanently lubricated type), matching gears, and all other moving parts installed in the lathe shall be provided means to assure adequate lubrication. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. All oil holes, grease fittings, and filler caps shall be so located as to be readily accessible. The supplier shall include and install all lubricants, greases and other lubricating compounds. Vent openings in the lubrication systems shall be temporarily capped or plugged to prevent loss of lubricants during shipment. Tags, warning that "ALL TEMPORARY CAPS OR PLUGS MUST BE REMOVED BEFORE

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START-UP OR USE OF THE ITEM", shall be attached to all temporary caps or plugs intended to be vented to the atmosphere during operation. These warnings shall also be included in the instruction manual. In the event a lubricant reservoir cannot be covered and sealed, it shall be appropriately preserved and the lubricant shall be provided in a sealed container and shipped with each piece of end item equipment. A warning tag shall be placed in a conspicuous place near the filling area, warning the user to "INSTALL THE FURNISHED LUBRICANT BEFORE OPERATION OF THE EQUIPMENT". The operator's manual shall also reflect this above warning and procedure.

3.2.7 Safety and health requirements. Each machine shall be furnished with safety devices. Parts which are hazardous to the operator shall be guarded. Protection against electric shock shall be provided. The machine shall comply with OSHA standards applicable to the machine itself.

3.2.8 Surfaces. Surfaces of all castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free of sand, dirt, fins, sprues, scale, flux and other harmful or extraneous materials.

3.2.9 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with best commercial practices, shall be suitable for the intended purpose in the design and fabrication of the lathe, and shall meet all requirements specified herein. Materials shall be free from defects which would affect the performance, maintainability, reliability, durability, or longevity of the individual components or the lathe. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.2.10 Raw materials. The manufacturer may use reclaimed materials for fabricating new parts. Reclaimed materials shall be reprocessed, remanufactured, or recycled in a manner that restores them to the same chemical composition and physical properties as the raw materials originally selected for use. Use of reclaimed parts as is or rebuilt from scrap or other used equipment shall not be permitted.

3.3 Fabrication. The lathe shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the lathe in its intended environment, to its maximum rating and capacity, without permanent distortion.

3.3.1 Castings and forgings. Finished castings and forgings shall be free of fins, scale, inclusions, cold shuts, voids, cracks, thermal ruptures, laps, folds, mismatching, etc. No processes such as welding, peening, plugging, soldering, or pasting shall be used for repairing, reworking, or improving cosmetic appearance.

3.3.2 Welding, brazing and soldering. All welding, brazing and soldering shall be free of defects and of a quality which shall sustain all requirements of the welded, brazed or soldered parts. Defects shall include but are not limited to such things as bubbles, cracks, gouges, holes, insufficient material used, etc. All flux or other such materials used during these operations shall be thoroughly removed (cleaned), from the part(s) immediately upon completion of the operation. These operations shall not be employed as repair measures for defective parts.

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3.4 Performance and product characteristics. The lathe shall be capable of truing commercial and military passenger car and truck brake drums and disc brake rotors within the sizes specified herein with one initial machine setup for each drum and rotor. The entire friction surface of the drum or rotor shall be trued without interrupting the rough and finish cuts to relocate drum or rotor on the spindle, relocate tool bar or holder, or relocate carriage or cross slide. Axial, horizontal spindle and arbor movement shall not be acceptable. The lathe shall be capable of truing brake drums and disc brake rotors weighing up to 200 pounds, with the wheels removed, within the sizes and to the accuracies specified herein without the aid of an outboard support. Performance accuracy shall apply after turning/truing the brake drum or disc brake rotor with the lathe, and shall be unaffected by the mounting position of the brake drum or disc on the arbor. Performance accuracy shall also apply with the brake drum or disc brake rotor mounted on a vehicle spindle. The drum or disc rotor shall be mounted with proper companion bearing assemblies (cups, cones and ball or roller bearings) in place and torqued in accordance with bearing manufacturer's recommendations. Turning and truing of the drums and rotors shall be accomplished without chatter. If a grinder or grinding attachment (see 3.5.10.2) is required to meet the required finishes specified in paragraphs 3.4.3.c and 3.4.4.c it shall be furnished with the lathe.

3.4.1 Capacities, capabilities, and characteristics. The lathe capacities, capabilities and characteristics shall be as follows. Where a range is indicated, the required performance is from the smaller figure or less to the larger figure or more.

a. Drum diameter range - 6 inch inside diameter to 24 inch inside diameter.
b. Drum width (friction surface) cutting range - 1 to 9 inches.
c. Drum machine powered (not hand powered) cutting feed rate range - 0.005 to 0.015 inches per revolution.
d. Number of machine powered (not hand powered) cutting feed rates for drums - Infinite or not less than 2.
e. Disc rotor friction surface cutting diameter range capacity - 4 inch inside diameter to 19-1/2 inch outside diameter.
f. Rotor width range (face to face of friction surfaces) - 0.125 to 1.75 inches.
g. Disc rotor machine powered (not hand powered) cutting feed rate range 0.002 to 0.015 inches per revolution.
h. Number of machine powered (not hand powered) cutting feed rates for rotors - Infinite or not less than 2.
i. Spindle speed range - 60 rpm to 115 rpm.
j. Number of machine powered spindle speeds - Infinite or not less than 3 (low speed, mid-range speed and high speed).

3.4.2 Design accuracy. The lathe and accessories shall not exceed the following tolerances:

a. Spindle taper run-out (concentricity) 0.0005 inch total indicator reading (TIR).
b. Arbor run-out (concentricity) 0.0005 inch TIR at all positions along its length up to 12 inches from the spindle nose and 0.001 inch TIR from 12 inches to 18 inches.
c. Adapters such as short radius cones, taper cones, sleeves, bushings, and spacers; run-out and concentricity to their true axis .0005 inch TIR.
d. Carriage alignment 0.001 inch TIR with the true axis of the spindle, at any point.

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The following degrees:

3.4.3 Performance accuracy for brake drums. The lathe shall be capable of truing all brake drums specified herein within the following tolerances:

a. Concentricity (turned surface to axis of hub). The turned surface of the brake drum shall be concentric with the axis of the bearing surfaces in the hub within 0.0015 inch TIR for brake drums up to 24 inches in diameter and friction surfaces up to 9 inches in depth. Applicable to rough cutting as well as finish cut.
b. Taper. The turned surface of the brake drum shall be true with the axis of the bearing surfaces of the hub within 0.0015 inch TIR up to 9 inches of brake drum friction surface and up to 24 inches in diameter. Applicable to rough cutting as well as finisn cut.
c. Finished surface. The final turned surface finisn of the brake drum shall not have a finish numerically greater than 40 micro-inches arithmetical average (AA).

3.4.3.1 Rough cutting for brake drums. The lathe shall rough cut brake drums of at least 24 inches inside diameter. The brake drum lathe shall also rough cut brake drums of at least 9 inch depth of friction surface. Rough cutting does not have to comply with finished requirements of para 3.4.3(c). These drums shall be rough cut as follows:

a. At not less than 60 RPM.
b. At not less than 0.025 inch deep cut (0.025 radius or 0.050 diameter).
c. At not less than 0.015 inches per revolution feed rate of the carriage tool bar and cutting tool.

3.4.4 Performance accuracy for disc brake rotors. The lathe shall be capable of simultaneously truing both friction surfaces of disc brake rotors, from 4 inch inside diameter to 19-1/2 inch outside diameter, within the following tolerances:

a. Each finished trued surface of the rotor shall not deviate from the true plane of the rotor surface in excess of ± 0.0005 inch or a TIR of 0.001 when measured face to face. The true plane of the rotor surface is a plane perpendicular to the axis of rotation which is established by the bearing surfaces of the hub and spindle assembly (see 3.4). Applicable to rough cutting as well as finish cut.
b. Thickness variations (parallelism) of the rotor between trued surfaces shall not exceed 0.001 inch. Applicable to rough cutting as well as finish cut.
c. The trued finished rotor surfaces shall not have a surface finish numerically greater than 40 micro-inches AA.

3.4.4.1 Rough cutting requirements for disc brake rotors. The brake drum and disc lathe shall rough cut the friction surface of brake disc rotors from 4 inch or less inside diameter to 19-1/2 inch or more outside diameter:

a. At not less than 60 RPM.
b. At not less than 0.015 inch deep cut on both sides of the disc rotor simultaneously.
c. At not less than 0.015 inches per revolution (IPR) feed rate of cross slide tool holder and cutting tool.

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3.5 Detail of components. The lathe shall consist of, but not be limited to, the following principal component parts.

3.5.1 Base. The floor mounting base shall be cast iron or heavy steel fabrication to absorb and prevent resonant vibration and to assure strength, rigidity and zero deformation when the lathe is operated under full load. Welded base structures shall be stress relieved. The base shall provide for permanent floor mounting (see 1.1) of the lathe with the center line of the lathe spindle at operator working height of no less than 35 inches from the floor and no more than 42 inches from the floor. Compartments shall be provided in the base for the storage of short radius cones, arbors, adapters, and tools required in the resurfacing of brake drums and disc brake rotors.

3.5.2 Headstock. The headstock housing shall contain the main spindle assembly and the matching ways for the carriage. The headstock housing shall be of a cast iron or cast steel construction, stress relieved, and shall be of sufficient strength to assure rigidity and zero deformation while operating under full load. The headstock assembly shall consist of, but not be limited to, the main spindle, gears, gear change mechanisms, feed change mechanisms, a handwheel drive for manually positioning the cutting tool, and the headstock ways. Torque transmitting headstock gearing shall be hardened, to at least Rockwell C50, ground steel gears.

3.5.2.1 Headstock ways. The headstock ways shall provide the bearing surface for the tool travel (parallel to the spindle) for the operation of turning and truing brake drums. The headstock ways shall be finished to provide an accurate uniform bearing surface for the matching carriage ways. The headstock ways shall be of sufficient length to maintain a full 100 percent load bearing contact, without any cantilever effect with the matching set of carriage ways, throughout the full carriage travel, while machining all specified sizes of brake drums. The headstock ways shall also be of adequate size and strength to maintain the required accuracies (see 3.4.2, 3.4.3 and 3.4.4).

3.5.3 Carriage. The carriage shall provide the matching bearing surface for carriage travel (parallel to the spindle) for the operation of turning and truing brake drums. If applicable, the carriage may also provide the matching bearing surface for the cross slide travel (perpendicular to the spindle) for the operation of turning and truing disc brake rotors. The carriage ways shall be of sufficient length to maintain a full 100 percent load bearing contact, without any cantilever effect with the matching set of headstock ways, throughout the full carriage travel, while machining all specified sizes of brake drums and disc rotors. The carriage shall be of a cast iron or cast steel construction, stress relieved, and shall be of sufficient strength to assure rigidity and zero deformation while operating under full load. The carriage shall consist of, but not be limited to the following:

a. A handwheel for manually positioning the cutting tool to the proximity of the work piece.
b. A feed dial graduated in increments to indicate 0.001-inch movements of the carriage.
c. Provisions for locking the carriage in all positions necessary for cutting.
d. A carriage adjustment gib to compensate for wear (see 3.5.3.2).
e. Carriage ways for the matching headstock and, if applicable, cross slide ways (see 3.5.3.1).

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3.5.3.1 Carriage ways. The carriage ways shall be finished to provide an accurate uniform bearing surface for the matching ways. The carriage ways shall be of sufficient length to maintain a full 100 percent load bearing contact, without any cantilever effect with the matching set of ways, throughout the full carriage travel and, if applicable, cross slide travel, while machining all specified sizes of brake drums and disc brake rotors. An automatic shut off device shall be incorporated in the carriage ways that will either disable the motor, or disengage the machine powered cutting feed mechanism in the event that the ways lose the full 100 percent load bearing contact with the matching set of ways.

3.5.3.2 Carriage gib. The carriage gib shall provide for a fine, precision adjustment to compensate for wear in the carriage ways, throughout the life of the lathe. The carriage gib shall provide an even pressure along the surface area of the ways.

3.5.4 Cross slide. The cross slide provides the matching bearing surface for the carriage ways (perpendicular to the spindle) for the operation of turning and truing disc brake rotors. The cross slide also provides the movable tool support for the turning and truing of disc brake rotors and, if applicable, brake drums. The cross slide shall be of a cast iron or cast steel construction, stress relieved, and shall consist of, but not be limited to the following:

a. A hand wheel and feed dial, graduated in increments to indicate 0.001-inch movements of the cross slide.
b. Provisions for locking the cross slide in any position.
c. Cross slide adjustment gib to compensate for wear (see 3.5.4.1).
d. Provisions for mounting the tool holder rigidly in its proper position on the cross slide.
e. Cross slide ways (see 3.5.4.2).

3.5.4.1 Cross slide gib. The cross slide gib shall provide for a fine, precision adjustment to compensate for wear in the cross slide ways throughout the life of the lathe. The cross slide gib shall provide an even pressure along the surface area of the ways.

3.5.4.2 Cross slide ways. The cross slide ways provide the mating bearing surface for the carriage ways (perpendicular to the spindle) for the operation of turning and truing disc brake rotors. The cross slide ways shall be of sufficient length to maintain a full 100 percent load bearing contact, without any cantilever effect with the matching carriage ways, throughout the full cross slide travel range while machining all specified sizes of disc brake rotors.

3.5.5 Tool bar. The lathe tool bar shall be made from steel and shall be designed to hold the cutting tool rigidly and securely, and prevent change in the rake angle of the cutting tool. The tool bar shall be designed so that it can be rotated 180 degrees in the clamp for turning drums with external braking surfaces.

3.5.6 Machine powered feed drive. There shall be two machine powered feed systems. One for machining brake drums, and the other for machining disc brake rotors. Each of the machine powered feed systems shall have either an infinite feed rate or not less than two feed rates, as specified in 3.4.1, one for rough machining cuts, and the other for finish machining cuts. Each machine powered feed system shall be provided with a clutch lever or a means of engaging the machine powered feeds. All feed rate changes, clutch levers, and controls shall be easily accessible to the operator and shall be easily adjustable. All half nuts, gears, clutches, etc. required in conjunction with the

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machine powered feed systems shall be enclosed. Feed screws and/or feed rods may be exposed.

3.5.6.1 Handwheels for manual positioning. Handwheels shall be provided for manually positioning the cutting tool to the proximity of the surface to be machined, pick up the initial cut, set the depth of cut, etc. There shall be a handwheel to provide manual movement of the carriage and another to provide manual movement of the cross slide.

3.5.6.2 Automatic stops. Automatic stop devices, for the prevention of damage to the equipment, for both grinding and turning, shall be as follows:

a. An automatic stop device shall be provided that will disengage the machine powered feed mechanism at a point that will prevent any contact between the carriage and the headstock.
b. An automatic stop device shall be provided for the cross slide ways and the carriage ways, as specified in 3.5.3.1, that will either shut off the motor or disengage the machine powered feed mechanism, to prevent any overhang or cantilever effect and to assure that the carriage or cross slide maintains the full 100 percent load bearing contact with the matching set of ways.

3.5.7 Spindles and arbors. Spindles and arbors shall be alloy steel, heat treated and ground on all aligning surfaces. Spindles shall be heat treated to a surface hardness of Rockwell C40-45. Arbors shall be heat treated to a surface hardness of at least Rockwell C50. Spindles and arbors shall be of a size to assure the accuracy required of the lathe and mounted to permit turning brake drums and disc brake rotors of the size specified. The arbors shall be mounted in the spindle by means of a suitable taper (from at least 0.5 inch per foot to not more than 6.5 inches per foot on diameter) with a positive lock and release system which will prevent damage to spindle or arbor. The length of the arbor shall be adequate to mount and surface finish cut the entire friction surface of the sizes of brake drums, disc brake rotors and nubs specified, in one set up and cut.

3.5.7.1 Spindle drive. A minimum of three spindle speeds or infinitely variable spindle speeds shall be provided. Gear drives shall be totally enclosed. Easily adjustable tensioning devices and guards shall be provided where belt or chain drives are employed. All necessary motor mounting brackets, pulleys, belts, chains, sprockets, gears, countershafts, and other parts required to successfully operate the lathe (as specified herein) shall be furnished. Visual means shall be provided to show spindle speeds throughout the entire speed range specified herein.

3.5.8 Electrical equipment.

3.5.8.1 Motors. The motors shall conform to the requirements of NEMA MG 1 Motors and Generators, as applicable. The machine shall be furnished with a continuous duty, induction squirrel-cage type, at least 2 horsepower motor with a temperature rise of not more than 50 degrees Centigrade above ambient temperature, thermal overload protected. Mounting shall be in accordance with NEMA standards. Motors shall be protected from dirt and grit and shall be equipped with sealed-for-life, antifriction bearings. A start-stop switch shall be mounted and wired, convenient to the operator.

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3.5.8.2 Three phase electrical system. The electrical system of the lathe shall operate from electrical power sources of 208 Volt, 230 Volt and 460 volts, 50 and 60 Hertz, three phase; initially wired to operate from 208/230 volts. A four wire (one for ground) electric power supply cable shall be installed on each piece of the lathe. The electric power supply cable shall be at least 10 feet in length and shall terminate in at least 1 inch of stripped tinned wire, without a receptacle plug. The electric power supply cable shall include proper sheathing/covering in accordance with the NFPA 70 and OSHA 29 CFR 1910. The electric power supply cable shall have a voltage/potential and ampere/current carrying capacity exceeding that required by the electric power consuming devices, including electric motors, in accordance with the NFPA 70. The electrical system shall conform to applicable requirements of NFPA 79. A disconnect switch isolating electric power shall be provided to disengage all incoming electricity, separate from the normal "ON" and "OFF" switch. An identified ground terminal shall be provided to connect the proper size grounding conductor or strap. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used, the conductors and terminal parts shall be mechanically secured before soldering. Only rosin base fluxes shall be used in soldering operations. Connections to screw type terminals shall be mechanically secured with a means to prevent loss of tightness.

3.5.8.3 Light fixture. The lathe shall be provided with a work light located in a position which will not interfere with controls, yet will afford full illumination at the cutting tool area. The light shall be equipped with a separate switch that shall function independently of the motor switch.

3.5.8.4 High voltage requirement. The entire electrical system of the end item shall withstand, without damage, malfunction, breakdown, arcing, surface or air discharge, at least 2500 volts peak, 60 Hertz, sinusoidal wave electrical energy at standard temperature and pressure. The electrical energy shall be applied instantaneously at full voltage and shall be maintained for at least 15 seconds. Each circuit shall be isolated from other circuits and all circuits shall be isolated from electrical ground. Solid state control components may be damaged as a result of high voltage and is not cause for rejection. Damage to solid state control components may be avoided by wiring around or by-passing them, terminal to terminal. No arcing or discharge from solid state control components is acceptable.

3.5.8.5 Coded electrical circuitry, wiring diagrams and schematics. The electrical circuitry used in the end item of equipment shall be accurately portrayed in electrical schematics and wiring diagrams furnished with each end item of equipment. Each individual wire and device used in the electrical circuitry of the actual hardware end item of equipment shall be permanently identified by insulation cover coloring; numbers, letters, or combinations thereof. The wiring diagram shall clearly and accurately portray the wiring, individual point to point wire terminations, and the electrical device connection points. Electrical schematics shall accurately portray the electrical system of the end item of equipment, showing all solid state devices, switches, relays, relay contacts, indicator lights, motors, etc., which are used in the system.

3.5.9 Accessories. All commercial and necessary accessories for turning and truing brake drums and disc brake rotors of military and commercial vehicles shall be furnished. For commercial and military vehicles, such accessories shall be provided to fit all bearing cups from the minimum size bearing cup Military Standards MS19081-18 thru the maximum size bearing cups MS19081-173 and MS19081-184 thru MS19081-313. Hardened steel, ground, short radius cones shall be furnished to fit all sizes and configuration of bearing cups. Collets are not acceptable. Tapered cone adaptors shall be provided for

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all commercial and military brake drums and discs to fully cover the sizes specified herein. Truck drum and disc anti-chatter devices shall be provided to cover the range of sizes specified herein.

3.5.10 Optional equipment.

3.5.10.1 Case. When specified, a suitable sturdy case of sheet metal or wood shall be furnished to contain the grinding attachment, gages, and all accessories for transportation of the equipment.

3.5.10.2 Grinding attachment. If a grinding attachment is required to meet the finishes specified in paragraphs 3.4.3.c and 3.4.4.c, it shall be furnished with the lathe. When specified, a grinding attachment shall be furnished with each lathe with six grinding wheels suitable for grinding cast-iron and/or hardened-steel brake drums, in accordance with ANSI B5.17, B7.1, and B46.1. A suitable guard, an abrasive-wheel dresser having a mounted octahedron diamond of 1/4-carat minimum size, feed mechanism, brackets, and necessary equipment to make a complete unit shall be included. The grinding attachment shall be capable of driving the abrasive wheel at a working surface speed of 3,600 to 7,000 feet per minute.

3.6 Fungus resistance. Electrical and electronic components and electrical insulation not otherwise specified herein shall be fungus inert, in accordance with requirement 4 of MIL-STD-454. Components of electrical circuitry that are not wherewith redistant to moisture and fungus growth shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.7 Painting and protective finish. Painting and protective finishing shall be in accordance with the best practice of commercial industry. The protective finish shall be a paint coating of at least 2.0 mils (0.002 inch) thickness. All surfaces to be painted shall, immediately prior to painting, be cleaned, free of all foreign matter, and dried. Paint shall adhere to the surface of the end item, when fiberglass tape is removed from two intersecting knife cuts, at least two inches long, through to the base metal. The surface hardness of the paint shall be between the industrial standards of H to HB pencil lead. Where no protective finish is utilized (such as bedways, machine guide surfaces, etc.) the natural finish of the material or the finish obtained from heat treatment is permissible provided the surfaces are free of scale and have a protective coating to inhibit corrosion.

3.8 Identification marking. The lathes shall be marked for identification in accordance with MIL-STD-130 and, unless otherwise specified, shall include the National Stock Number.

3.9 Nameplate. The nameplate and information plate(s) can be incorporated into one or more corrosion-resistant metallic plate(s) at the machine manufacturer's option. Characters and numerals shall be permanently and legibly engraved, etched, embossed, or stamped in bold face on a contrasting background. The plate shall be secured to the machine with permanent fasteners such as screws, bolts, or rivets at each of the four corners. The plate shall include the following information:

Nomenclature
National Stock Number (NSN)
Manufacturer's name
Manufacturer's model number
Manufacturer's serial number (if applicable)

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Contract number

3.10 Instruction and information plates. Corrosion-resistant metallic information and instruction plates with permanently legible characters, printed in the American-English language, shall be permanently fastened to the end item in locations visually convenient to the operator. Characters and numerals shall be permanently and legibly engraved, etched, embossed, or stamped in bold face on a contrasting background. The plates shall be secured to the end item with permanent fasteners such as screws, bolts, or rivets at each of the four corners of the plates. The information plate(s) shall include warnings, cautions, and machine operator instructions to be followed when operating or maintaining the end item.

3.11 Lubrication chart or plate. A corrosion-resistant metallic nameplate, with permanently legible characters, written in the American-English language, shall be attached to each end item of equipment in a location conveniently visible to the operator. If a chart is furnished, it shall be placed in a transparent plastic folder or laminated between clear plastic, permanently sealed sheets with suitable means for mounting. The following information shall be furnished on the chart and plate:

Points of application
Service interval
Type of lubricant
Viscosity
Military or Federal specification of lubricant

3.12 Workmanship. The workmansnip employed in the manufacture of the end item of equipment shall be equal to or exceed the quality of commercial items manufactured by major producers of the equipment as specified herein. In any instance where the requirements herein exceed those of the commerical item counterpart, the workmanship shall meet or exceed the requirements herein. The consistency of the workmanship shall be such as to eliminate degredation of the specified performance, design reliability. Workmanship displayed in the fabrication and assembly shall be such as to assure requirements herein are met. The items and its components shall be constructed and finished to produce an item free from all defects which would affect proper functioning of the item in service and provide for operator safety. Workmanship criteria shall include, but not be limited to, neatness and thoroughness of the structural integrity and structual soundness of:

a. Assemblies b. Soldering c. Welding d. Brazing

e. Painting f. Riveting g. Machine-screw assemblies

h. Freedom of parts from exhibiting poor material and processing such as:

(1) Seams (2) Laps (3) Laminations (4) Cracks

(5) Visible steps (6) Sharp edges and corners (unless necessary to perform a required function)

(7) Nicks (8) Scratches (9) Burrs

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(10) Deformations and missing operations which affect ability, functioning, operations, appearance or safety.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that the products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1)
(b) Quality conformance inspection (see 4.2.2)

4.2.1 First article inspection. When specified, first article inspection shall be performed on the preproduction model(s) or initial production item(s) noted in 3.1. First article inspection shall consist of the examination in 4.3, all tests in 4.4, and the packaging inspection in 4.5. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.2.2 Quality conformance inspection.

4.2.2.1 Inspections and tests. Quality conformance inspection shall be applied to production units offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (j) as follows, and failure of any unit to pass an examination or test shall be cause for rejection of the unit.

(a) Product examination (see 4.3)
(b) Machine accuracy test (see 4.4.1)
(c) Drum truing test (see 4.4.2)
(d) Disc truing test (see 4.4.3)
(e) Automatic stop test (see 4.4.5)
(f) High voltage requirement test (see 4.4.6)
(g) Grinding wheel speed test (see 4.4.7)
(h) Paint adhesion test (see 4.4.8)
(i) Hardness tests (see 4.4.9)
(j) Packaging inspection (see 4.5)

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4.2.2.2 Sampling. Sampling for quality conformance inspection shall be performed in accordance with MIL-STD-105, using the following.

Inspection or test	Level	AQL
Product examination (4.3)	II	1.0
Machine accuracy test (4.4.1)	II	1.0
Drum turning test (4.4.2)	S-4	1.0
Disc truing test (4.4.3)	S-4	1.0
Automatic stop test (4.4.5)	S-4	1.0
High voltage req'm't test (4.4.6)	S-4	1.0
Grinding wheel speed test (4.4.7)	S-4	1.0
Paint adhesion test (4.4.8)	S-4	1.0
Hardness tests (4.4.9)	S-4	2.5
Packaging inspection (4.5)	S-4	4.0

4.3 Product examination. The lathes shall be visually, manually, and dimensionally examined to determine conformance with the requirements of 3.2 through 3.3.2, 3.4.1, and 3.5 through 3.12 (including 3.5.10.1 and 3.5.10.2 if applicable). Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. Dimensional examination shall verify size and weight conditions when specified. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the lathes and their components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.2.1), shall be cause for rejection.

4.4 Tests. Measuring instruments used in performing the following tests shall be more accurate by a factor of ten than the specified limit, and shall be traceable to the National Bureau of Standards. Failure to pass any of the test requirements shall be cause for rejection of the lot.

4.4.1 Machine accuracy test. The spindle taper run-out, arbor run-out, adapter run-out and concentricity, and carriage alignment of the lathe shall be measured. Failure of the machine to comply with the requirements of 3.4.2 shall be cause for rejection.

4.4.2 Drum truing test. Three brake drums shall be set up and turned as a typical operation. A large brake drum (minimum diameter 24 inches) and a small brake drum (maximum diameter 6 inches) shall be turned, as well as a wide brake drum (minimum friction surface width 9 inches). The large brake drum shall weigh at least 200 pounds and shall be trued without the aid of an outboard support. (At the contractor's option, weight may be attached to the brake drum as necessary to meet the weight requirement.) Performance accuracy of the lathe shall be determinined by measuring the taper and concentricity of the turned surface to the spindle with dial indicators accurate within ±0.0001 inch, while the freshly turned drum is still mounted on the lathe. The drum shall then be loosened on the arbor, rotated to 120 and 240 degrees from its original position on stationary arbor and short radius cone, and measured for concentricity with

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the spindle in both positions in the same manner as above. For the first article test only, the turned drum shall then be mounted on a vehicle spindle with proper companion bearing assemblies (cups, cones, and ball or roller bearings) in place and torqued in accordance with the bearing manufacturer's recommendations, and re-measured for taper and concentricity with the vehicle spindle. The final finish of the turned drum shall then be checked with a surface analyzer or profilometer. Failure of the lathe to attain the performance requirements of 3.4 and the rough cut, final finish, and accuracy requirements of 3.4.3 and 3.4.3.1 shall be cause for rejection.

4.4.3 Disc truing test. Two brake discs shall be set up and trued as a typical operation. A large brake disc (maximum friction surface diameter at least 19 inches) and a small brake disc (minimum friction surface diameter not more than 5 inches) shall be trued. The large brake disc shall weigh at least 200 pounds. (At the contractor's option, weight may be attached to the disc as necessary to meet this requirement.) Performance accuracy of the lathe shall be determined by measuring the true plane of each rotor surface and the parallelism of the rotor surfaces with dial indicators accurate within +0.0001 inch, while the freshly trued disc is still mounted on the lathe. The disc shall then be loosened on the arbor, rotated to 120 and 240 degrees from its origional position on the stationary arbor and snort radius cones, and the true planes and parallelism of the rotor surfaces re-measured in both positions in the same manner as above. For the first article test only, the trued disc shall then be mounted on a vehicle spindle with proper companion bearing assemblies (cups, cones, and ball or roller bearings) in place and torqued in accordance with the bearing manufacturer's recommendations, and the true planes and parallelism of the rotor surfaces re-measured. The final finish of the trued disc shall then be measured with a surface analyzer or profilometer. Failure of the lathe to attain the performance requirements of 3.4 and the rough cut, final finish, and accuracy requirements of 3.4.4 and 3.4.4.1 shall be cause for rejection.

4.4.4 Spindle speed test. The lathe shall be tested with a tachometer to determine speeds and speed ranges. Failure of the spindle speeds to comply with 3.4.1.(i) and 3.4.1.(j) shall be cause for rejection.

4.4.5 Automatic stop test. Without a brake disc or brake drum mounted on the lathe, the lathe shall be turned on and the carriage and cross-slide drives operated to the limits of their travel. Failure of the automatic stops to operate in accordance with 3.5.6.2 shall be cause for rejection.

4.4.6 High voltage requirement test. A 60 Hertz test voltage shall be applied between machine ground and an uninsulated point on each isolated circuit in the machine. The volt-ampere cpability of the test voltage source shall be of a sufficient capacity to maintain a voltage of at least 2500 volts peak when applied to a nondefective circuit. The test voltage shall be applied to each isolated circuit for no less than 15 seconds. All ON-OFF switches and circuit breakers shall be in the ON position when performing this test. Damage to solid state components may be avoided during this test by removing or shunting these components. Damage to solid state components is not cause for rejection, but all damaged or degraded components must be replaced before acceptance by the Government. Any damage to any component other than solid state, or evidence of malfunction, breakdown, arcing, or other surface or air discharge, or a drop in test voltage below 2500 volts peak at any time during the test, shall be cause3 for rejection.

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4.4.7 Grinding wheel speed test. When a grinding attachment is specified in the contract, the spindle speed of the grinding attachment shall be measured with a tachometer while performing grinding operations on the brake drums tested in 4.4.2. The test shall include use of each size and type of grinding wheel provided with the attachment. Failure of the attachment to maintain a calculated surface speed of 3,600 to 7,000 feet per minute for each wheel during the grinding operation shall be cause for rejection (see 3.5.10.2).

4.4.8 Paint test. The hardness, adhesion and thickness of the paint used on the machine shall be checked as follows:

(a) The grapnite point of a sharpened HB lead pencil shall be pressed at a right angle into a painted area of the unit until the graphite crumbles. Failure of the surface to withstand the test, as evidenced by a pitting of the paint, shall be cause for rejection.

(b) A test sample of the same metal, painted with the same process and at the same time as the unit, shall have a sharpened H lead pencil pressed into the paint as in (a). Failure of the surface to pit (paint too hard) shall be cause for rejection.

(c) Two, two-inch intersecting lines shall be cut through the paint to the base metal of the test sample. A strip of 1/2 inch wide fiberglass tape shall be firmly pressed over the intersection and then pulled from the surface. Adhesion of any paint to the tape shall be cause for rejection.

(d) Using a paint thickness measuring instrument, or by masking a portion of the test sample during painting, measure the thickness of the paint. Failure of the paint to be at least 0.002 inch thick shall be cause for rejection.

4.4.9 Hardness tests. The lathe spindles and arbors shall be hardness tested to Rockwell C, or equivalent superficial scales may be used and converted to Rockwell C. Spindle hardness shall be in the Rockwell C40 to C45 range and arbor hardness shall be at least Rockwell C50 (see 3.5.7). Headstock gearing hardness shall be at least Rockwell C50 (see 3.5.2).

4.5 Packaging inspection. The lathes shall be inspected before and after packaging to determine compliance with the preservation, packaging, packing, and marking requirements specified in Section 5 of this specification.

5 PACKAGING

5.1 First article. When specified, the supplier shall furnish one complete pack for first article packaging inspection. The first article may be either a preproduction pack or an initial production pack which conforms to the requirements stated herein. In either case, the approved first article and the production packs shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements stated herein.

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5.2 Disassembly. When practical, for the protection of components, attachments and accessories against damage, pilferage or to reduce cubage, items may be disassembled only to a necessary basic degree. Disassembled parts shall be preserved, anchored, braced, blocked, and cushioned to prevent damage. Disassembled parts shall be clearly and legibly marked as to identity and proper location on the assembled item. All fasteners removed during disassembly shall be secured in one of the mating parts. A part shall not be removed from an assembly unless it can be reassembled readily in the field without special skills or tools.

5.3 Disassembled parts marking. Disassembled parts shall be marked, when necessary with instructions, to facilitate reassembly. Removed parts and mating parts on the item shall be marked by stenciling, on the part, or by the use of tags. Tags shall be waterproofed and ink used (on parts or tags) shall be waterproof. Tags shall be firmly secured to each individual part.

5.4 General requirements. Component parts of machinery, accessories, and maintenance tools shall be cleaned, dried, preserved, packaged, and packed as specified herein. Parts and assemblies which are not covered herein shall be preserved and packaged utilizing the materials and methods specified for similar parts and assemblies described herein. Cleaning and drying procedures, and preservatives shall conform to the criteria and guidelines specified herein.

5.5 Preservation. Preservation shall be level A or C as specified.

5.5.1 Level A.

5.5.1.1 Cleaning and drying. Thorough cleaning and drying of all bare ferrous metal surfaces is essential before the application of a preservative. Care must be taken to limit the handling of cleaned parts prior to the assembly of new machinery. Where practicable, processors and assemblers shall wear rubber or plastic gloves while handling cleaned parts.

5.5.1.2 Internal surfaces. All ferrous metal surfaces of parts which will be utilized internally in the assembly of new machinery shall be solvent cleaned followed by fingerprint remover and wiped or drained dry. In the interim, to preclude contamination from forming on cleaned surfaces prior to the assembly operation, parts shall be coated with a light weight preservative oil conforming to P-9 of VV-L-800. To prevent dust from accumulating on cleaned parts, cover with a sheet of chemically neutral paper.

5.5.1.3 External surfaces. All accessories and attachments such as chucks, faceplates, steady rests, toolholders, abrasive wheels, etc. shall be removed from the assembly prior to the external cleaning operation. All external, critical ferrous metal surfaces, fit or function, of the machine shall be solvent cleaned followed by fingerprint remover and wiped or drain dried. Noncritical ferrous metal surfaces shall be solvent cleaned and wiped or drain dried. Nonmetallic composites and painted metal surfaces shall be cleaned by wiping with a clean, dry cloth. Care must be taken to prevent cleaning solvents from coming in contact with nonmetallic components. When cleaned surfaces of the machine show signs of rusting before application of the final preservative, reprocessing the rusting surface shall be confined to that area of the assembly. The solvent used for the final wiping operation shall contain a small amount, approximately 5 percent by volume, preservative oil conforming to P-10, type 10, grade 30 of MIL-L-21260. It is extremely important that elapsed time between these operations be held to an absolute minimum.

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has passed qualification tests as prescribed by either of the following listed

5.5.1.4 Disassembled components. Disassembled components detached from the basic machine snall be cleaned and dried as specified herein for like items.

5.5.1.5 Machine tools and accessories. All such tools and accessories shall be clean and free from contaminants. Preservatives shall be applied to provide protection from corrosion during shipment and storage. All preserved items shall be wrapped with greaseproof barrier material. Critical surface items, i.e. parallels, etc., shall also be wrapped with no less than 1/2 inch uncompressed cushioning material.

5.5.1.6 Electrical equipment. Electrical equipment shall be clean and dry. Preservatives shall be applied as required to prevent corrosion to moving or other ferrous metal parts, but shall not come into contact with electrical connections, windings, etc.. Openings shall be sealed witn cap-plugs or with waterproof, pressure sensitive tape to provide protection against entry of any foreign materials.

5.5.1.7 Lubricating systems. Vent openings in the lubrication systems shall be temporarily capped or plugged to prevent loss of lubricants during shipment. Tags, warning that "ALL TEMPORARY CAPS OR PLUGS MUST BE REMOVED BEFORE START- UP OR USE OF THE ITEM", shall be attached to all temporary caps or plugs intended to be vented to the atmospnere during operation. These warnings shall also be included in the instruction manual. In the event a lubricant reservoir cannot be covered and sealed, it shall be appropriately preserved and tne lubricant shall be provided in a sealed container and shipped with each machine. A warning tag shall be placed in a conspicuous place near the filling area, warning the user to "INSTALL THE FURNISHED LUBRICANT BEFORE OPERATION OF THE EQUIPMENT". Thc operator's manual shall also reflect tne above warning and procedure.

5.5.1.8 Maintenance and hand tools. Maintenance and hand tools such as screwdrivers, hammers, chisels, etc., shall be clean and free from contaminants. Sharp edges such as cutting edges shall be protected by a fiberboard shield placed on the edge and taped in place. Tools shall be placed in a fiberboard box and secured from movement to prevent damage to tne tools or other items.

5.5.1.9 High-speed spindle bearings. Fill the reservoir to normal operating level with manufacturer's operating lubricant. If the system is of such design that the oil will leak out during shipment or storage, tne system shall be preserved by filling with preservative oil P-10, type I, grade 30 of MIL-L-21260 and drained. Operational lubricant shall be furnished.

5.5.1.10 Grease fittings. Charge all grease fittings with preservative compound conforming to P-11 of MIL-G-10924.

5.5.1.11 Nonlubricated interior surfaces. Surfaces of internal gears which do not run in lubricant shall be thoroughly coated with P-2 of MIL-C-16173, grade 2 preservative.

5.5.1.12 Gear cases. Fill tne gear cases with normal lubricating oil. Shift gears into all possible positions while operating tne machine, then close and seal all openings.

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5.5.1.13 Sliding or friction surfaces. Surfaces of ways, gibs, driving gears, traversing and adjusting screws, and other critical, exterior operating machined surfaces shall be coated with P-2 of MIL-C-16173, grade 2 preservative compound and covered with greaseproof barrier material. Parts of the machine which are adjustable shall be locked into place.

5.5.1.14 Bearings and journals. Grease lubricated bearings shall be preserved with P-11 of MIL-G-10924 preservative. Oil lubricated journals shall be filled with lubricating oil P-10, type I, grade optional of MIL-L-21260.

5.5.1.15 Drive belts and pulleys. Drive belts shall either be removed from the machine or released from tension. Ferrous metal pulleys shall be coated with rust-inhibiting primer. Belts removed from the machine shall be packaged in a waterproof, greaseproof bag and sealed.

5.5.1.16 Electrical wiring system. Exposed ends of wires, plug openings, sockets, connector plugs, terminals, and openings into switches and junction boxes, etc. shall be sealed with waterproof tape or cap-plugs, respectively. Electric cable shall be coiled to a minimum safe diameter and secured with cotton tape or twine to prevent uncoiling.

5.5.1.17 Closed dial indicators. Closed dial indicators shall be covered with 1 inch thick uncompressed cushioning material and secured in place with waterproof, pressure sensitive tape. Cushioning materials and tape used shall be water-resistant, non-dusting and non-corrosive.

5.5.1.18 Drawers and door assemblies. Sliding surfaces of drawer guides shall be coated with P-11 of MIL-G-10924, preservative grease. Interior surfaces of door hinges shall be lubricated by application of P-10, grade 30 of MIL-L-21260, preservative oil. In addition to mechanical locks or catches incorporated in the machine, secure doors and drawers with metal strapping or other devices which are suitable for the purpose intended. Fiberboard pads shall be placed under metal strapping to prevent marring the surfaces.

5.5.1.19 Closure of openings. Small openings, except vents and louvers installed for ventilation purposes, shall be sealed with waterproof, pressure sensitive tape. Large openings shall be covered with waterproof barrier material. Secure materials in place with tape specified above. When very large openings are covered, or when location of openings render coverings vulnerable to puncture, coverings shall be protected by applying wood, plywood, sheet metal or fiberboard over the materials. Open ends of all piping and fittings shall be plugged to prevent the entrance of foreign materials, using pipe fittings, waterproof tape or plastic cap-plugs.

5.5.1.20 Technical data. Technical data furnished with the machine shall be packaged in a waterproof, greaseproof barrier material and sealed. The package shall be legibly and clearly marked "TECHNICAL DATA" in bold letters. The package shall be protected against puncture and abrasion. The data shall be either attached to the machine with tape, placed in a stowage compartment of the machine, or positioned in a consolidated box containing accessories or disassembled components.

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5.5.1.21 Consolidated packaging. Packaged accessories, maintenance tools, and small disassembled components shall be consolidated and placed in drawers and stowage compartments of the machine. When it is not feasible to stow parts as specified above, place parts in weather-resistant fiberboard box or boxes of sufficient strength to handle weight inside. Large bulky disassembled components shall be packed as specified (see 5.6.1).

5.5.1.22 Cushioning. Cushioning materials shall be water resistant, non-dusting and non-corrosive and of sufficient thickness to prevent damage, not less than 1/2 inch thick wrap.

5.5.1.23 Protection of preservatives. All exposed preserved surfaces of machines which are subject to being disturbed during the packing operation shall be loosely covered with greaseproof barrier material. Not less than two layers of similar-type barrier material shall be used to separate wood or metal blocking and finished metal surfaces of the machine. Secure barrier in place with waterproof, pressure sensitive tape.

5.5.1.24 Movable parts. Operating heads, sliding tables, rams, counter-balances, and other movable parts shall be secured by mechanical devices incorporated in the machine. The devices alone do not always provide adequate support to prevent movement in the handling and transporting of the item and therefore must be augmented with appropriate wood blocking, metal tie rods used in conjunction with wood blocking, metal strapping, or other devices which will immobilize the part. Handwheels, cranks, levers, and similar controls shall be secured in position with common annealed wire or steel strapping.

5.5.1.25 Positioning assemblies. Assemblies having vertical movement shall be blocked and braced in a lowered position; those capable of moving traversely shall be centered, blocked and braced to prevent movement.

5.5.1.26 Tables. All tables and other components designed for movement shall be removed, or blocked, and all components treated in such a manner that neither the way(s), surface(s), nor the anti-friction devices will be subject to brinelling or other damage.

5.5.2 Level C. Level C preservation shall be in accordance with MIL-STD-1190.

5.6 Packing. Packing shall be level A or C as specified.

5.6.1 Level A.

5.6.1.1 Machinery. Machines shall be packed individually in crates conforming to MIL-C-104, type I, class 2, style A. Consolidated containers shall be packed with the machine in the specified crate. Arrangement of contents, anchoring, blocking, bracing, and cushioning shall be in accordance with the appendix to the crate specification, MIL-STD-1186, and as specified herein. Consolidated containers for large bulky disassembled items (see 5.5.25) shall conform to PPP-B-601, domestic, style optional.

5.6.1.2 Bolting. Machines shall be secured to the skid base by not less than four bolts. When sufficient bolting points are not available, J-clamps or tie rod assemblies, or other devices, shall be used to provide the minimum of four holding points. Bolts shall pass completely through the skid runner or sleeper, as applicable, and the base flooring.

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5.6.1.3 Consolidated containers. Consolidated containers shall be placed inside the shipping container housing the machine. The boxes shall be restrained by use of metal strapping or tie rod assemblies.

5.6.1.4 Operational oils. Operational oils furnished with the machine, but not placed in the machine stowage receptacles, shall be stowed in metal drum(s). Drums shall be positioned upright inside the shipping container housing the machine. Blocking shall be applied to immobilize the drum(s) inside the shipping container.

5.6.2 Level C. Level C packing shall be in accordance with MIL-STD-1190.

5.7 Marking. All marking shall be in accordance with MIL-STD-129. Bar coding is also required in accordance with MIL-STD-129.

5.8 Workmanship. Workmanship shall be such that when proper procedure is followed, materials and equipment being processed will be provided the maximum protection against corrosion and deterioration consistent with the shipment hazards known or anticipated, and suitable for long term storage.

6 NOTES

6.1 Intended use. The floor mounted, with base, lathes covered by this specification are intended for use in truing brake drums and disc brake rotors of the commercial and military type.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. First Article, when required (see 3.1).

c. Case, if required (see 3.5.10.1).

d. National Stock Number marking, if not required (see 3.8).

e. Responsibility for inspection, if different (see 4.1).

f. First article inspection, if different (see 4.2.1).

6.3 Data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract (see 3.11). Repair parts and accessory identification in repair part manuals should include the prime contractor's name, nomenclature and part number. Repair part identification in repair part manuals should also include the original equipment/part manufacturer's (OEM) (actual part manufacturer's) name, Contractor and Government Entity Code (CAGE), nomenclature and part number of the part used in the end item, immediately after the prime contractor's part identification. Items with high mortality rates should be so identified.

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6.4 Subject term (key word) listing.
Drum Lathe
Rotor Lathe

Custodian:
Army-AL

Preparing Activity:
Army-AL
Project No:
4910-A569

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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